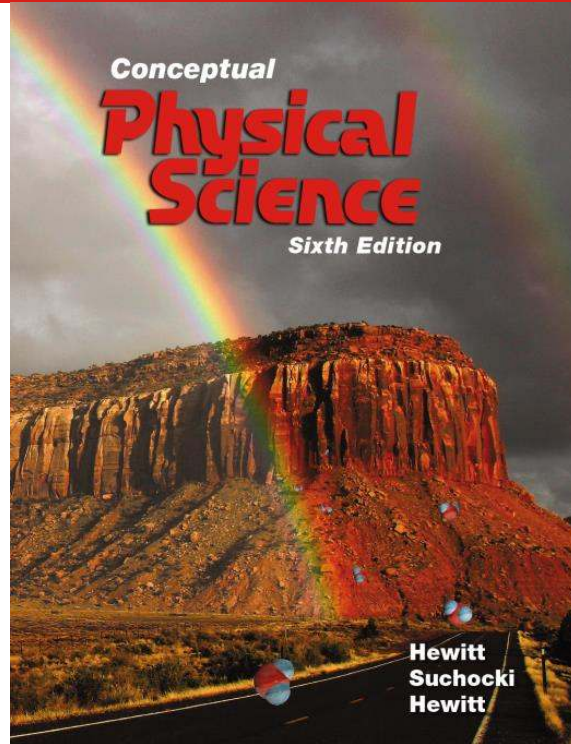


Chapter 4: Gravity, Projectiles, and Satellites

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The Universal Law of Gravity

- Newton discovered that gravity is *universal*.
- Every mass pulls on every other mass.

- Check the cavendish experiment
<https://youtu.be/MbucRPiL92Q?t=450>



x any two masses in
universe attract each
other

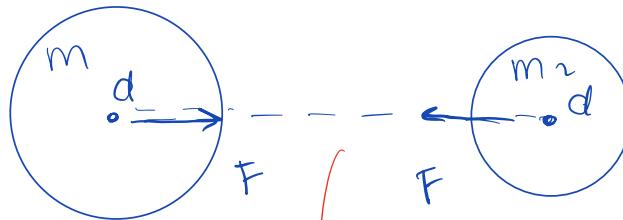


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newtons law of gravitation

$m \uparrow \quad F \uparrow$
 $d \uparrow \quad F \downarrow$



$$F = \frac{G m_1 m_2}{d^2}, \quad G = 6.67 \times 10^{-11}$$

$\downarrow$ N $\uparrow$ kg $\downarrow$ m

$\rightarrow$ 2 on 1, 1 on 2
 have the same magnitude due to action reaction

x derive unit of G

$$N = \frac{G \text{ kg} \times \text{kg}}{m^2} = \frac{x \text{ kg}^2}{m^2} = N$$

$$\frac{N m^2}{\text{kg}^2} = \frac{\text{kg}^2 \times}{\text{kg}^2}$$

$$x = \frac{N m^2}{\text{kg}^2}$$

$$F = \frac{G m_1 m_2}{d^2}$$

$m \uparrow \quad F \uparrow$
 $d \uparrow \quad F \downarrow$

m_1	m_2	d	F	
$2m_1$	m_2	d	$2F$	
$3m_1$	$5m_1$	d	$3 \times 5 F$	$= 15F$
m_1	m_2	$2d$	$\left(\frac{1}{2}\right)^2 F$	$= \frac{1}{4} F$
m_1	m_2	$\frac{d}{4}$	$\left(\frac{4}{1}\right)^2 F$	$= 16F$
$\frac{1}{4}m_1$	$16m_2$	$\frac{3}{2}d$		

$\rightarrow \frac{1}{4} \times 16 \times \left(\frac{2}{3}\right)^2 F$

m_1	m_2	d	F
$3m_1$	$\frac{1}{9}m_2$	$3d$	$3 \times \frac{1}{9} \times \left(\frac{1}{3}\right)^2 F = \frac{1}{27} F$
$5m_1$	$10m_2$	$\frac{2d}{5}$	$5 \times 10 \times \left(\frac{5}{2}\right)^2 F = \frac{625}{2} F$
$\frac{1}{2}m_1$	$\frac{1}{4}m_2$	$8d$	$\frac{1}{2} \times \frac{1}{4} \times \left(\frac{1}{8}\right)^2 F = \frac{1}{512}$
$\frac{1}{9}m_1$	$3m_2$	$\frac{1}{6}d$	$\frac{1}{9} \times 3 \times (6)^2 F = 12F$

The Universal Law of Gravity

- Newton's Law of Universal Gravitation
 - Every body in the universe attracts every other body with a mutually attracting force.
 - For two bodies, this force is directly proportional to the product of their masses and inversely proportional to the square of the distance separating them,

$$F = G \frac{m_1 m_2}{d^2}$$

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The Universal Gravitational Constant, G

- G is the proportionality constant in Newton's law of gravitation.
- G has the same magnitude as the gravitational force between two 1-kg masses that are 1 meter apart:

$$6.67 \times 10^{-11} \text{ N}.$$

$$\text{So } G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2.$$

$$F = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2 (m_1 \times m_2)/d^2$$

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Newton's Law of Universal Gravity

$$F = G \frac{m_1 m_2}{d^2}$$

- The greater m_1 and $m_2 \Rightarrow$ the greater the force of attraction between them.
- The greater the distance of separation d , the weaker is the force of attraction—weaker as the inverse square of the distance between their centers.

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The Inverse-Square Law: Gravity and Distance

CHECK YOUR NEIGHBOR

The force of gravity between two planets depends on their

- A. masses and distance apart.
- B. planetary atmospheres.
- C. rotational motions.
- D. All of the above.

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The Inverse-Square Law: Gravity and Distance

CHECK YOUR ANSWER

The force of gravity between two planets depends on their

- A. masses and distance apart.
- B. planetary atmospheres.
- C. rotational motions.
- D. All of the above.

Explanation:

The equation for gravitational force,

$$F = G \frac{m_1 m_2}{d^2}$$

cites only masses and distances as variables. Rotation and atmospheres are irrelevant.

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The Inverse-Square Law: Gravity and Distance

CHECK YOUR NEIGHBOR

If the masses of two planets are each somehow doubled, the force of gravity between them

- A. doubles.
- ☒ B. quadruples.
- C. reduces by half.
- D. reduces by one quarter.

$$F = G \frac{m_1 m_2}{d^2}$$

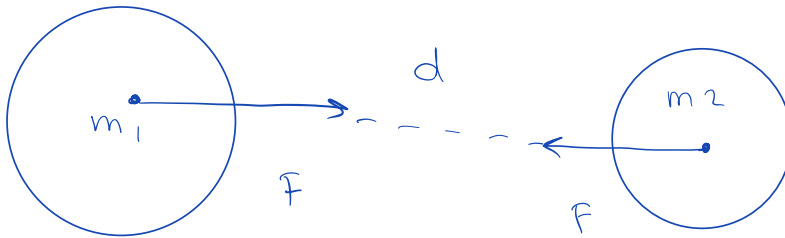
$$F = G \frac{2m_1 2m_2}{d^2}$$

$$F = G \frac{4m_1 m_2}{d^2}$$

$$F = 4F$$

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$$F = G \frac{m_1 m_2}{d^2}$$

$\nearrow \text{kg}$
 $\nwarrow \text{N}$

whereby $G = 6.67 \times 10^{-11}$

The Inverse-Square Law: Gravity and Distance

CHECK YOUR ANSWER

If the masses of two planets are each somehow doubled, the force of gravity between them

- A. doubles.
- B. quadruples.**
- C. reduces by half.
- D. reduces by one quarter.

Explanation:

Note that both masses double, so double $\times$ double = quadruple.

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The Universal Law of Gravity

CHECK YOUR NEIGHBOR

If the mass of one planet is somehow doubled, the force of gravity between it and a neighboring planet would

- A. doubles.**
- B. quadruples.
- C. reduces by half.
- D. reduces by one quarter.

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The Universal Law of Gravity

CHECK YOUR ANSWER

If the mass of one planet is somehow doubled, the force of gravity between it and a neighboring planet would

- A. doubles.
- B. quadruples.
- C. reduces by half.
- D. reduces by one quarter.

Explanation:

Let the equation guide your thinking:

$$F = G \frac{m_1 m_2}{d^2}$$

Note that if one mass doubles, the force between them doubles.

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Example

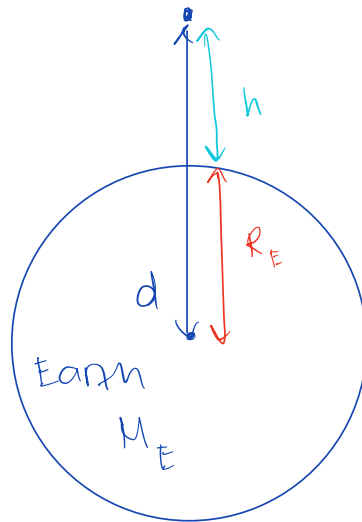
- Note: Gravitational force is weak when masses are small.

Example: Calculate the gravitational attractive force between two people of masses 90 kg and 50 kg when they are 4 m apart.

$$F = G \frac{m_1 m_2}{d^2} \quad \text{whereby } m_1 = 90 \text{ kg}, m_2 = 50 \text{ kg}, d = 4 \text{ m},$$
$$G = 6.67 \times 10^{-11}$$

$$F = (6.67 \times 10^{-11}) \frac{(90)(50)}{(4)^2} = 1.8759 \dots \times 10^{-8}$$
$$\approx 1.88 \times 10^{-8} \text{ N}$$

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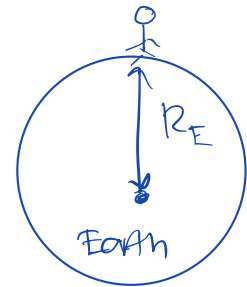
$$d = R_E + h$$

$$F = \frac{G M_E \cdot m}{(R_E + h)^2}$$

Example

Calculate the gravitational attractive force between an 80 kg man and Earth when he is at earth surface. Given that mass of Earth is 6×10^{24} kg and the radius of Earth is 6380 km.

$$\begin{aligned}
 m &= 80 \text{ kg} & G &= 6.67 \times 10^{-11} \\
 M_E &= 6 \times 10^{24} \text{ kg} \\
 R_E &= 6380 \times 10^3 \text{ m} \\
 h &= 0 \\
 F &= (6.67 \times 10^{-11}) \frac{(80)(6 \times 10^{24})}{(6380000)^2} \\
 &= 786.54 \\
 &\approx 787 \text{ N}
 \end{aligned}$$



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The Universal Law of Gravity

- Note that the gravitational force in the previous example is equal to the weight of the man when calculated using $w = mg$ (with small difference due to approx. values)
- We can write

$$W = m_{\text{man}} g = G \frac{m_{\text{Earth}} m_{\text{man}}}{d^2}$$



- Therefore:

$$g_{\text{Earth}} = G \frac{m_{\text{Earth}}}{R_{\text{Earth}}^2}$$

- Where d is set to equal the radius of Earth (R_{Earth})
- The above formula can be used to calculate the acceleration of gravity on another planet, given its radius and mass

$$F = \frac{G M_E \times \cancel{m}}{(R_E + h)^2} = \cancel{m} \cdot g$$

$$g_{\text{Earth}} = \frac{G M_E \rightarrow \text{kg}}{(R_E + h)^2}$$

$\swarrow \frac{\text{m}}{\text{s}^2}$
 $\searrow \text{m}$

$$g_{\text{planet}} = \frac{G M_{\text{planet}} \rightarrow \text{kg}}{(R_{\text{planet}} + h)^2}$$

$\swarrow \frac{\text{m}}{\text{s}^2}$
 $\searrow \text{m}$

Calculate the gravitational attractive force between a 90 kg man and Earth when he is above the surface of Earth at h. Given that mass of Earth is 6×10^{24} kg and the radius of Earth is 6380 km.

$$m = 90 \text{ kg} / M_E = 6 \times 10^{24} \text{ kg}$$

$$R_E = 6380 \text{ km}$$

$$= 6380 \times 10^3 \text{ m}$$

$$h = 5250 \text{ m}$$

$$d = R_E + h$$

$$= (6380 \times 10^3) + 5250 = 6385250 \text{ m}$$

$$G = 6.67 \times 10^{-11}$$

$$F = G \frac{m \cdot M_E}{(R_E + h)^2}$$

$$F = (6.67 \times 10^{-11}) \frac{(90)(6 \times 10^{24})}{(6385250)^2}$$

$$F = 883.4 \text{ N}$$

$$m + M_E$$

$$90 + (6 \times 10^{24})$$

$$h + R_E$$

$$5250 + 6380$$

$$F = G \frac{m \cdot M_E}{(h + R_E)^2}$$

Example

gravitational force = weight

Example:

a) Find the weight of a 70 kg man when he is at (2500 km) above earth surface.

$$\begin{aligned} \cdot G &= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2} & \cdot h &= (2500 \text{ km}) \times 10^3 \\ \cdot m &= 70 \text{ kg} & F &= \frac{(6.67 \times 10^{-11})(70)(6 \times 10^{24})}{((6380 \text{ km} \times 10^3) + (2500 \text{ km} \times 10^3))^2} = 355.26 \text{ N} \\ \cdot M_E &= 6 \times 10^{24} \text{ kg} \\ \cdot R_E &= (6380 \text{ km} \times 10^3) \text{ m} \end{aligned}$$

b) use your answer to part (a) to calculate the acceleration of gravity at that height. (in two different methods)

$$w = mg$$

$$F = ma \text{ / } W = mg$$

$$\frac{355 \text{ N}}{70} = \frac{(70)}{70} a \quad , \quad a = 5.07 \text{ m/s}^2$$

2nd method

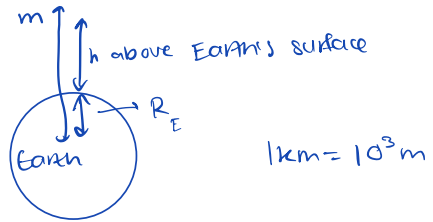
$$g = \frac{GM_E}{(R_E + h)^2} = \frac{(6.67 \times 10^{-11})(6 \times 10^{24})}{((6380 \times 10^3) + (2500 \times 10^3))^2}$$

$$5.07 \text{ m/s}^2 \approx 5 \text{ m/s}^2$$

(acceleration of gravity)

Summary

$$F = \frac{G m_1 m_2}{d^2}$$



$$F = \frac{G m_E \times \cancel{m}}{(R_E + h)^2} = \cancel{m} g$$

$$g_E = \frac{G m_E}{(R_E + h)^2} \quad , \quad g_{\text{planet/star}} = \frac{G M_{\text{planet}}}{(R_{\text{planet}} + h)^2}$$